II nd Shift
18/05/2017

Paper Code- AHE4081

SET-B

GLA University

Soft Skills (Even Sem)

End Term Examination (2016-17)

Course- B.Tech Year- II

Branch- All Branches

Instructions :

1. There are 2 sections in the paper.
2. Attempt all questions. The maximum marks for the test are 40.
3. The total time allocated is 120 minutes.
4. There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.
5. The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.
6. The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.
7. Only black& blue pen should be used to fill forms.
8. Each branch is identified with a specific subject code which should be written in subject column of OMR sheet
9. Write your University roll number in the roll no column of OMR sheet.

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:.....

Signature of Invigilator-.....

S.N	Branch	Code
1	CS	1500
2	CE	1000
3	ECE	1300
4	EN	1400
5	EE	1100
6	ME	1200

LIFE@GLA

Section-1 (Reasoning Ability)- No of Questions-40, marks-20

Q(1/5)-During a trial, an attorney is planning to call to the stand to testify six different witness- T, V, W, X, Y and Z. She will call each witness one time, and the order in which she calls them is subject to the following conditions: The fourth witness must be T. W and Y must both be called before V. X must be called before W.

Q1. Which of the following could be the order in which the attorney calls the six witnesses?
A. WXYTZV B. XWVTYZ C. XYWTZV D. ZTXWYV E. ZXWTVY

Q2. If the attorney calls Z last which witness must be called fifth?
A. T B. V C. W D. X E. Y

Q3. If the attorney calls X immediately after y and immediately before Z, she must call V?
A. first B. second C. third D. fifth E. sixth

Q4. The attorney could call any of the following witnesses immediately before T , except
A. V B. W C. X D. Y E. Z

Q5. If the attorney calls Z first, which of the following could be her fifth and sixth witnesses respectively?
A. T and X B. V and W C. X and V D. Y and V E. Y and X

Directions Q(6-10):
In the following questions the symbol $, @, *, ** and # are used with the following meaning.
A $ B means A is greater than B
A @ B means A is either greater than or equal to B
A * B means A is equal to B
A ** B means A is smaller than B
A # B means A is either smaller than or equal to B
Now in each of the following questions assumingthe given statements to be true, find whichof the two conclusions I and II given below them is/are definitely true?
Give answer
A.if only conclusion I is true.
B.if only conclusion II is true.
C.if either I or II is true.
D.if neither I nor II is true.
E.if both I and II are true.

Q6. **Statements :** P @ Q, M # N, N**Q
Conclusions :
I. P $ M
II. N # P

Q7. **Statements :** D**X, F @ Y, D $ F
Conclusions :
I. X @ Y
II. Y # D

Q8. **Statements :** M**P, S $ T, M @ T
Conclusions :
I. S * M
II. T ** P

LIFE@GLA

Q9. **Statements :** U*V, X $ W, U**W
Conclusions :
I. W $ V
II. U ** X

Q10. **Statements :** G $ H, J # K, H * K
Conclusions :
I. H $ J
II. J @ H

Q(11/15)-Study the following information to answer the given questions.
A word and number arrangement machine when given an input line of words and numbers rearrange them following a particular rule. The following is an illustration of input and rearrangement. (Single digit numbers are preceded by a zero. All other numbers are two digit numbers)

Input : whether 15 wish you 08 being 48 come 68 hour 124 86
Step I. : being whether 15 wish you 08 48 come 68 hour 124 86
Step II. being 08 whether 15 wish you 48 come 68 hour 124 86
Step III. being 08 come whether 15 wish you 48 68 hour 124 86
Step IV. being 08 come 15 whether wish you 48 68 hour 124 86
Step V. being 08 come 15 hour whether wish you 48 68 124 86
Step VI. being 08 come 15 hour 48 whether wish you 68 124 86
Ste VII. being 08 come 15 hour 48 whether 68 wish you 124 86
Step VIII. being 08 come 15 hour 48 whether 68 wish 86 you 124

Step VIII is the last step of the arrangement of the above input as the intended arrangement is obtained.As per the rules followed in the above steps, find out in each of the following questions the appropriate steps for the given input.

Q(11/14)-Input : new 59 personnel 68 28 teacher 10 price grievance 32

Q11.How many steps would be needed to complete the arrangement?
A. V B. VI C. VIII D. VII E. CND

Q12. Which of the following would be the final arrangement?
A. grievance 10 new 28 personnel 32 price 59 68 teacher
B. grievance new personnel price teacher 10 28 32 59 68
C. grievance 10 new 28 personnel 32 price 59 teacher 68
D. grievance 68 new 59 price 32 personnel 28 new 10
E. None of these

Q13. Which of the following would be step I?
A. grievance new 59 68 personnel 28 teacher 10 price 32
B.10 grievance new 59 personnel 68 28 teacher price 32
C. grievance 10 new 59 personnel 68 28 teacher price 32
D. 10 grievance new 59 personnel 68 28 teacher price 32
E. grievance new 59 personnel 68 28 teacher 10 price 32

Q14. Which word/number would be the sixth position from the left end in step III ?
A. personnel B. price C. 68 D. 32 E. teacher

Q15.**Input : jam 14 aim virtue 22 trouble 5 15**
Which word/number would be at position 5 from the right end in step III ?
A. aim B. 15 C. jam D. 14 E. trouble

(Q. 16 - 18) : Read the following information carefully and answer the questions given below it.
1. Six flats on a floor in two rows facing north and south are alloted to A, B, C, D, E and F.
2. B gets a south facing flat and is at one of the end.
3. D and F are diagonally opposite flats.
4. E, next to F, gets a North facing flat and C gets a south facing flat.

Q16. Which of the following combinations gets North facing flats?
A.AEF B.ACD C.BCD D.EDC

Q17. Whose flat is between B and D?
A.F B.E C.A D.C

Q18. If the flats A and F are interchanged, Whose flat will be opposite to that of B?
A.E B.C C.A D.D

Q19.In a row of boys, If A who is 10th from the left and B who is 9th from the right interchange their positions, A becomes 15th from the left. How many boys are there in the row ?
A. 23 B. 31 C. 27 D. 28

Q20. In a class of 180, where girls are twice the number of boys, Rupesh [a boy] ranked 34th from the top. If there are 18 girls ahead of Rupesh, how many boys are after him in rank?
A. 45 B. 44 C. 60 D. can't be determined

Q21. Mr. 'Ashok' meets Mr.'Babu'. 'Babu' is the father of a son 'Dharmendra' and a daughter 'Chandrika'. 'Shalini' is the mother of 'Ashok' 'Dharmendra' is married has one son. 'Shalini' is the daughter-in-law of 'Babu'. How is 'Ashok' related to 'Babu'.
A. Grandson B. Nephew C. Uncle D. Son E. NOT

Q22. If Krish says Chetan's sister is my father's only daughter which of the following is true about the relationship of Krish and Chetan?
A. Krish is Chetan's nephew B. They are real brothers C. They are cousins
D. Krish is Chetan's father E. None of these

Q(23/25)-Directions : Read the following information to answer the given questions.
(i) A, B, C, D, E, F and G are playing cards sitting around a circular table
(ii) D is not neighbour of C or E.
(iii)A is neighbour of B and C.
(iv) G, who is second to the left of D, is the neighbour of E and F.

Q23. Which of the following is correct?
A. B is between A and D B. D is between F and G.
C. E is to the immediate right of G D. F is to the immediate left of G

Q24. Which of the following has the second person sitting between the first and third person?
A. DAB B. BCA C.GFD D. GCE

Q25. Which of the following is wrong?
A. A is to the immediate right of B B. B is to the immediate left of D
C. F is between G and D D. E is between G and C

Q26. If PRIVATE is coded as **1234567** and **RISK** is coded as **2398**, how is **RIVETS** coded?
A. 232679 B.243769 C. 234769 D. 234976

Q27. In a certain code, **SIKKIM** is written as **THLJJL**, how is **TRAINING** written in that code ?
A. SBQHOHOF B. UQBOHIOF C. UBQHOHOI D. UQBHOHOF

Q28. If **JOSEPH** is coded as **FKOALD**, then **GEORGE** will be coded as :
A.HAKNCA B. CBKNCA C. CALNCA D. CAKNCA

Q29. If **ENGLAND** is written as **1234526** and **FRANCE** is written as **785291,** how is GREECE coded ?
A. 381171 B. 381191 C. 832252 D. 835545

Q30. If white is called blue, blue is called red, red is called yellow, yellow is called green, green is called black, black is called violet and violet is called orange, what would be the colour of human blood ?
A. Red B. Green C. Yellow D. Violet E. Orange

Q31. If the animals which can walk are called swimmers, animals who crawl are called flying, those living in water are called snakes and those which fly in the sky are called hunters, then what will a lizard be called ?
A. Swimmers B. Snakes C. Flying D. Hunters E. None of these

Q(32/36)-A cube of 14 cm has been painted on its surfaces in such a way that 2 opposite surfaces have been painted blue & two adjacent surfaces have been painted red. Two remaining surfaces have been left unpainted. Now the cube is cut into smaller cubes of side 2 cm each.

Q32.How many cubes will have no side painted?
A.125 B.216 C.175 D.180

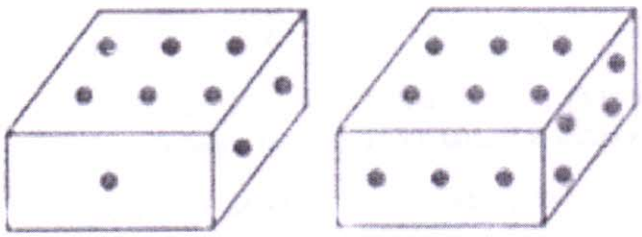
Q33. How many cubes will have at least red colour on its surface?
A.98 B.91 C.97 D.92

Q34. How many cubes will have at least blue colour on its surfaces?
A.90 B.98 C.94 D.93

Q35. How many cubes will have only two surfaces painted with red & blue, respectively?
A.8 B.12 C.24 D.30

Q36. How many cubes will have three surfaces painted?
A.3 B.4 C.2 D.16

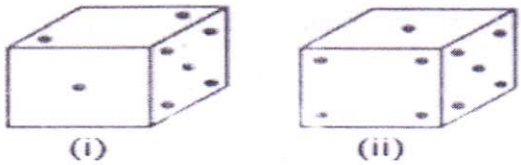
Q37. Two positions of a block are shown below: When six is at the bottom, what number will be at the top?



(i) (ii)

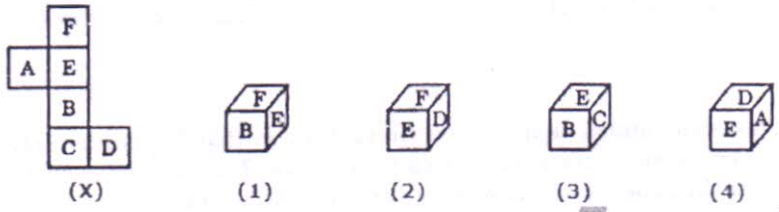
A.3 B.4 C.5 D.1

Q38.Observe the dots on a dice (one to six dots) in the following figures. How many dots are contained on the face opposite to that containing four dots?



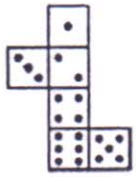
A.3 B.4 C.2 D.1

Q39. Choose the box that is similar to the box formed from the given sheet of paper (X).



A.1 only B.2 only C.1 and 3 only D.1, 2, 3 and 4 only

Q40. When the following figure is folded to form a cube, how many dots lie opposite the face bearing five dots?



A.1 B.2 C.3 D.4

Section-2 (Quantitative Aptitude)- No of Questions-40, marks-20

Q41. 270416X is divisible by 13. Find X

A. 1 B. 9 C. 7 D. None of these

Q42. The population of a town increased by 12.5% during 1st year and decreased by 10% during second year. If the present population is 16,200, find the initial population.

A. 16,000 B. 20,000 C. 12,000 D. None of these

Q43. A shopkeeper sells an article at a gain of 10%. Had he sold it at a loss of 20%, its selling price would have been Rs. 180 less. What is the cost price of the article?

A. Rs. 630 B. Rs. 610 C. Rs. 580 D. None of these

Q44. A sum of 1521 is lent out on 2 parts on such a way that the interest on one part @ 10% for 5 years is equal to that on another part @8% for 10 years. Find the two sums.

A. Rs. 639, Rs. 882
B. Rs. 875 and Rs. 646
C. Rs. 936, Rs. 585
D. None of these

Q45. Rs.100 doubled in 5 years when compounded annually. How many more years will it take to get another Rs.200 compound interest?

A. 10 years
B. 5 years
C. 7.5 years
D. None of these

Q46. A and B started a business in partnership investing Rs. 20,000 and Rs. 15,000 respectively. After six months, C joined them with Rs. 20,000. What will be B's share in total profit of Rs. 25,000 earned at the end of 2 years from the starting of the business?

A. 7,500
B. 9,000
C. 9,500
D. None of these

Q47. The retail price of a water geyser is Rs. 1265. If the manufacturer gains 10%, the wholesale dealer gains 15% and the retailer gains 25%, then the cost of the product is

A. Rs. 800
B. Rs. 900
C. Rs. 700
D. None of these

Q48. Find the average of all numbers between 6 and 34 which are divisible by 5

A. 15
B. 30
C. 25
D. None of these

Q49. Average weight of 10 people increased by 1.5 kg when one person of 45 kg is replaced by a new man. Then weight of the new man is

A. 50
B. 55
C. 60
D. None of these

Q50. A vendor sells 60 percent of apples he had and throws away 15 percent of the remainder. Next day he sells 50 percent of the remainder and throws away the rest. What percent of his apples does the vendor throw?

A. 17
B. 23
C. 77
D. None of these

Q51. If Rs. 782 is divided into three parts, proportional to $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$ find the first part.

A. 190
B. 204
C. 220
D. None of these

Q52. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is

A. 2:5
B. 3:5
C. 4:5
D. None of these

Q53. A dog takes 3 leaps for every 5 leaps of a hare. If one leap of the dog is equal to 3 leaps of the hare, the ratio of the speed of the dog to that of the hare is

A. 9:5
B. 4:5
C. 8:5
D. None of these

LIFE@GLA

Q54. A writer gets a fixed amount for his book apart from the royalty that he gets per book sold. He gets Rs. 30000 and Rs. 50000 for 1000 and 2000 books sold respectively. What is his income per book, if 5000 books are sold?

A. Rs. 17 B. Rs. 20 C.Rs. 21 D. None of these

Q55. A 20 litre mixture of milk and water contains milk and water in the ratio 3 : 2. 10 litres of the mixture is removed and replaced with pure milk and the operation is repeated once more. At the end of the two removals and replacement, what is the ratio of milk and water in the resultant mixture?

A. 17 : 3 B. 9 : 1 C. 3 : 17 D. None of these

Q56. One quantity of wheat at Rs 9.30 per Kg is mixed with another quality at a certain rate in the ratio 8:7. If the mixture so formed be worth Rs 10 per Kg, what is the rate per Kg of the second quality of wheat?

A. Rs. 12.47 B. Rs. 15.17 C. Rs. 47.66 D. None of these

Q57. A milk vendor has 2 cans of milk .The first contains 25% water and the rest milk. The second contains 50% water. How much milk should he mix from each of the container so as to get 12 litres of milk such that the ratio of water to milk is 3:5?

A. 6 litres B. 1 litres C. 8 litres D. None of these

Q58. An alloy contains zinc, copper and tin in the ratio 2:3:1 and another contains copper, tin and lead in the ratio 5:4:3. If equal weights of both alloys are melted together to form a third alloy, then the weight of lead per kg in new alloy will be:

A. 1/10 kg B. 1/8 kg C. 1/4 kg D. None of these

Q59. The first and the last digit of a three digit number differ by four. Find the difference of the number and the number formed by reversing the digits?

A. 396 B. 463 C. 242 D. None of these

Q60. If a positive number is decreased by 2 and the squared the resultant number is 2 less than the original number. Find a possible value of the number.

A. 1 B. 2 C. 3 D. None of these

Q61. The cost of three pens, four rulers and five refills is Rs. 75 while that of ten refills, six pens and seven rulers is Rs. 138. Find the cost of three pens, one ruler and five refills.

A. Rs. 39 B. Rs. 42 C. Rs. 44 D. Cannot be determined

Q62. A is two years older than B who is twice as old as C. If the total of the ages of A, B and C be 27, the how old is B?

A. 7 B. 8 C. 9 D. None of these

Q63. If $5 + \log_{10} x = 5 \log_{10} y$, then express x in terms of y.

A. $x = 5y^5$ B. $x = (y/10)^5$ C. $x^2 = 10y^5$ D. None of these

Q64. N is a 15-digit number. Find the integral part of $\log_{10} N$

A. 14 B. 15 C. 16 D. None of these

Q65. If $\log_{81} 25 = k \log_3 5$ find k.

A. 2 B. 4 C. $\frac{1}{2}$ D. None of these

Q66. Log $\frac{16}{15}$ + 5 Log $\frac{25}{24}$ + 3 Log $\frac{81}{80}$ = Log x, x = ?

A. 2 B. 3 C. 0 D. None of these

Q67. The LCM & HCF of 8/9, 32/81 & 10/27 ?

A. 160/9, 2/81 B. 160/12, 2/8 C. 180/9, 2/9 D. None of these

Q68. The least number which when divided by 5, 6, 7, or 8 leaves the remainder 3 in each case, but when divided by 9 leaves no remainder is

A. 1677 B. 1683 C. 1697 D. None of these

Q69. The least number which should be added to 3500 to make the result exactly divisible by 42, 49, 56 and 63 is

A. 28 B. 56 C. 3528 D. None of these

Q70. If $\sqrt[3]{32} = 2^x$, then x is equal to:

A. 5 B. 3 C. $\frac{3}{5}$ D. None of these

Q71. If $2^{x-1} + 2^{x+1} = 320$, then value of x is :

A. 6 B. 8 C. 5 D. None of these

Q72. The value of $\frac{3^{(12 \cdot n)} \times 9^{(2n-7)}}{3^{5n}}$ is :

A. $\frac{1}{3}$ B. $\frac{9}{13}$ C. $\frac{1}{9}$ D. None of these

LIFE@CLA

Q73. Find the eighth term of a geometric progression whose first term as well as common ratio is 2.

A. 128 B. 256 C. 512 D. None of these

Q74. Find the greater of the two numbers whose arithmetic mean is 20 and whose geometric mean is 16.

A. 18 B. 20 C. 24 D. None of these

Q75. Find the maximum value of n for which the sum to n terms of the progression 7, 10, 13, ... is less than 2500.

A. 35 B. 38 C. 39 D. None of these

Q76. The 69th term of an arithmetic progression is 16 times the fourth term of that arithmetic progression. Find the 20th term of the arithmetic progression, if its 7th term is 22.

A. 61 B. 48 C. 53 D. None of these

Q77. Find the sum to n terms of the series

$$\log_2 2, \log_2 4, \log_2 8, \log_2 16 \ldots\ldots$$

A. $\frac{n(n+1)}{2}$ B. $\frac{n(2n+1)(n+1)}{6}$ C. $\left[\frac{n(n+1)}{2}\right]^2$ D. None of these

Q78. At present the ratio between the ages of Jai and Vijay is 4: 3. After 6 years Jai's age will be 26 years. What is the age of Vijay at present?

A. 12 years B. $19\frac{1}{2}$ years C. 15years D. None of these

Q79. If $2^{2x-1} = \frac{1}{8^{x-3}}$, the n the value of x is:

A. 3 B. 2 C. 0 D. None of these

Q80. If $(9)^3$ is subtracted from the square of a number, the answer so obtained is 567. What is the number?

A. 36 B. 28 C. 42 D. None of these

LIE@GLA

10/5/17 1st Shift

SET-A

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), End Term Examination, 2016-17

Subject: English for Professional Purposes–II Subject Code: AHE 4083

Time: 90 Minutes Max. Marks: 90

VERBAL APTITUDE

Identify the correct part of speech of the underlined word from given options. (Q. 1-7)

1. His naturalistic behaviour appealed to all the panelists.
   A. Adjective B. Noun C. Pronoun
   D. Adverb E. None of these
2. Because my friend was quite occupied, he couldn't take my call.
   A. Conjunction B. Preposition C. Pronoun
   D. Noun E. Adverb
3. How did Kattapa kill Bahubali?
   A. Noun B. Preposition C. Conjunction
   D. Adjective E. Adverb
4. The utterly delicious dinner didn't cost me a fortune.
   A. Adjective B. Verb C. Noun
   D. Adverb E. None of these
5. The unusually high pricing of the things couldn't deter him from making that purchase.
   A. Adjective B. verb C. preposition
   D. Adverb E. Noun
6. Sleepwalking is a kind of disease which is very rare in India.
   A. Present participle B. Past Participle C. Gerund
   D. Infinitive E. None of these
7. A calculator is nothing but a computing device.
   A. Present participle B. Past Participle C. Gerund
   D. Infinitive E. None of these

Identify the correct option to fill in the blank. (Q. 8-10)

8. _____ or not to marry is no question to me.
   A. Marrying B. marry C. married
   D. to merry E. to marry
9. My friends didn't want _____the party so early.
   A. Leaving B. left C. leave
   D. to leave E. Both C & D
10. Entire South-East Asia was in a _____ state when Obama discovered Osama.
    A. Horrified B. horrific C. horrifying
    D. to horrify E. Both A & B

Use appropriate idiom/phrase to complete the sentence. (Q.11-15)

11. You know me, I _____and when I see someone behaving like an idiot, I tell them.
    A. Call a spade a spade D. Make a splash
    B. Make a spectacle of myself E. None of these
    C. Vent my spleen
12. We are standing on the _____of environment collapse.
    A. On the side of caution D. On the threshold
    B. On the side of angels E. None of these
    C. On the sidelines
13. It's no use asking Samishtha for help, she has_____looking after the kids.
    A. Kept her hands clean D. Got out of hand
    B. Got her hands full E. None of these
    C. Got a handle on
14. He was highly respected during his lifetime but he became a _____after his death when his research was found to be full of errors.
    A. Fair weather friend D. Fallen idol
    B. False dawn E. None of these
    C. False alarm
15. The lawyer did her best to _____in the witness's statement.
    A. Be holier than thou D. A home buddy
    B. Pick holes E. None of these
    C. Be Honest to goodness

Fill in the blank with suitable option. (Q. 16-35)

16. If you want to join that club, you have to _____ a form and send it to your secretary.
    A. fill up B. fill out C. fill into
    D. fill over E. fill through
17. What you have to do is to _____ a new idea that will enable us to make a lot of money.
    A. think over B. think out C. think about
    D. think up E. think in
18. It's easy to see from the way the garden is looking that winter has _____.
    A. set out B. set in C. set up
    D. set off E. set on
19. We need to _____ the price of the product, which is relatively high and focus on its quality as a selling point.
    A. back down B. break down C. play down
    D. settle down E. put down
20. I like that magazine, but I think we should _____ advertising in it until its circulation has increased.
    A. put out B. put back C. put away
    D. put off E. put up
21. The dead trees and peeling paint, along with the broken windows and flapping shutters, _____ everyone believe that evil spirits haunt the deserted Sinclair house.
    A. Made B. has made C. makes
    D. Make E. None of the above
22. Where _____ the earrings that I left by the bathroom sink? Oh no! That stupid cat has swatted them down the drain again!
    A. Was B. were C. is
    D. Are E. None of the above
23. Neither of those sharks circling your boogie board _____ hungry enough to bite.
    A. Looks B. look C. looked
    D. had looked E. None of the above
24. One hundred and fifty gallons _____ the amount of liquid the average living room rug can absorb. Rachel discovered this fact after taking a long shower when the curtain wasn't tucked inside the tub.
    A. Is B. are C. was
    D. Were E. None of the above
25. Agnes never loses a single possession. Everyone knows what belongs to her, for each pen, pencil, and paperclip _____ a tiny flag attached with Agnes' full name on it.
    A. will have B. has C. have
    D. had E. None of the above
26. CUP: LIP:: BIRD : _____?
    A. Bush B. Grass C. Forest
    D. Beak E. Thorn
27. Each of the following questions has a group. Find out which one of the given alternatives will be another member of the group or of that class.
    Basic, Pascal, Fortran, _____
    A. Cobol B. Bhopal C. Calculator
    D. Cyclotron E. Modulator

28. NEIGHBOR: PROXIMITY:: _____?
    A. supervisor: obedience
    B. comrade: order
    C. adversary: opposition
    D. traitor: distrust
    E. Mercenary: benign
29. CONDESCENDING: RESPECT :: _____?
    A. bashful: attention
    B. insecure: doubt
    C. merciless: compassion
    D. pathetic: pity
    E. rash : mild
30. COMPREHEND : UNINTELLIGIBLE:: _____?
    A. swallow : edible
    B. circumvent: risky
    C. accumulate : insignificant
    D. control : ungovernable
    E. accuse: blame
31. SEDATIVE : SLEEP:: _____?
    A. challenging : knowledge
    B. mnemonic : memory
    C. dramatic : story
    D. vocal: eloquence
    E. anaesthesia: nauseating
32. HATCH : HOLD :: _____?
    A. rudder: anchor
    B. boat: barge
    C. courtyard : terrace
    D. door: room
    E. plane: cockpit
33. INDIGENT: WEALTHY :: _____?
    A. angry: rich
    B. native: affluent
    C. gauche: graceful
    D. scholarly: erudite
    E. ace: excellent
34. AFTER: BEFORE :: _____?
    A. first: second
    B. present: past
    C. contemporary: historic
    D. successor: predecessor
    E. frontier: anterior
35. DISTANCE: MILE :: _____?
    A. liquid: litre
    B. bushel: corn
    C. weight: scale
    D. fame: television
    E. age: height

Identify the most suitable form of Indirect Speech. (Q. 36-40)

36. Robert will say to me, "I am your classmate".
    A. Robert will tell me that he is my classmate.
    B. Robert will tell me that he was my classmate.
    C. Robert will tell me that he will be my classmate.
    D. Robert said me that he is my classmate.
    E. None of the above
37. Sushant said to him, "I have been helping your son for years".
    A. Sushant told him that he has helped his son for years.
    B. Sushant told him that he have been helping his son for years.
    C. Sushant told him that he had been helping his son for years.
    D. Sushant told him that he has been helping his son for years.
    E. None of the above
38. My father said to me, "It might rain yesterday".
    A. My father told me that it may have rain the previous day.
    B. My father told me that it would rain the previous day.
    C. My father told me that it might rain the previous day.
    D. My father told me that it could rain the previous day.
    E. None of the above
39. Baljider said to me, "I had been working on it for 5 days."
    A. Baljinder told me that he had been working on it for 5 days.
    B. Baljinder told me that he has been working on it for 5 days.
    C. Baljinder told me that he had worked on it for 5 days.
    D. Baljinder told me that he was working on it for 5 days.
    E. None of the above
40. I said to my brother, "What are you playing today?"
    A. I asked my brother what he will be playing that day.
    B. I asked my brother what he was playing that day.
    C. I asked my brother if he was playing that day.
    D. I asked my brother what he would be playing that day.
    E. None of the above

Pick the most appropriate option. (Q. 41-45)

41. A. If I would live in a hill station, I would be much healthier.
    B. If I lived in a hill station, I would be much healthier.
    C. If I live in a hill station, I would be much healthier.
    D. If I will live in a hill station, I would be much healthier.
    E. None of the above.
42. A. Since that journey, I never sailed again.
    B. Since that journey, I have never been sailed again.
    C. Since that journey, I have never sailed again.
    D. Since that journey, I never sail again.
    E. None of the above.
43. A. He constantly got into trouble.
    B. He was constantly getting into trouble.
    C. He has constantly get into trouble.
    D. He had constantly got into trouble.
    E. None of the above.
44. A. I'm writing in reply to your letter that that I've received two days ago.
    B. I'm writing in reply to your letter that that I had received two days ago.
    C. I'm writing in reply to your letter that that I was received two days ago.
    D. I'm writing in reply to your letter that I received two days ago.
    E. None of the above.
45. A. My name is Ramu and I born in Bangladesh.
    B. My name is Ramu and I have been born in Bangladesh.
    C. My name is Ramu and I was born in Bangladesh.
    D. All of the above.
    E. None of the above.

Read the passages and answer the questions that follow. (Q. 46-55)

The happy man is the man who lived objectively, who has free affection and wide interest, who secures his happiness through these interests and affections and through the fact that they, in turn, make him an object of interest and affection to many others. To be the recipient of affection is a potent cause of happiness, but the man who demands affection is not the man upon whom it is bestowed. The man who receives affection is, speaking broadly, the man who gives it. But it useless to attempt to give it as a calculation, in the way in which one might lend money at interest, for a calculated affection is not genuine and is not felt to be so by the recipient.

What then can a man do who is unhappy because he is encased in self? So long as he continues to think about the cause of his unhappiness, he continues to be self-centred and therefore does not get outside, the vicious circle if he is to get outside it, it must be by genuine interests, not by simulated interest accepted merely as a medicine. Although this difficulty is real, there is nevertheless much that he can do if he has rightly diagnosed his trouble. If, for example, his trouble is due to a sense of sin, conscious or unconscious he can first persuade his conscious mind that he has no reason to feel sinful, and then proceed, to plant his rational

conviction in his unconscious mind, concerning himself meanwhile with some more or less neutral activity. If he succeeds in dispelling the sense of sin, it is possible that genuine objective interests will arise spontaneously. If his trouble is self-pity, he can deal with it in the same manner after first persuading himself that there is nothing extraordinarily unfortunate in his circumstances.

If fear is his trouble, let him practice exercises designed to give courage. Courage has been recognised from time immemorial as an important virtue, and a great part of training of boys and young men has been devoted to producing a type of character capable of fearlessness in battle. But moral courage and intellectual courage have been much less studied, they also, however, have their technique, admit to yourself every day at least one painful truth, your will find his quite useful. Teach yourself to feel that life still be worth living even if you were not, as of course you are immeasurably superior to all your friends in virtue and in intelligence. Exercises of this sort prolonged through several years will at last enable you to admit facts without flinching and will, in so doing, free you from the empire of feat over a very large filed.

46. If a man is suffering from a sense of sin.....
    A. He should invite opinion of others
    B. He should admit his sin at once
    C. He should consciously realize that he has no reason to feel sinful
    D. He should develop a fearless character
    E. He should develop an internal locus of control
47. What happens to a man who demands affection?
    A. His feelings are reciprocated by others
    B. He tends to take a calculated risk
    C. He becomes a victim of a vicious circles
    D. He takes affection for granted from others
    E. None of these
48. What should a man do who is suffering from the feeling of self-pity?
    A. He should control his passions and emotions
    B. He should persuade himself that everything is alright in his circumstances
    C. He should seek affection from others
    D. He should develop a feeling of fearlessness
    E. He should consult an expert to diagnose his trouble
49. Which of the following, according to the passage, has not been studied much?
    A. Feeling of guilt and self-pity
    B. The state of mind of a unhappy man
    C. How to get absorbed in other interests
    D. Moral and intellectual courage
    E. None of these
50. Which of the following words is SIMILAR in meaning of the word 'bestowed' as used in the passage?
    A. Conferred
    B. Accommodated
    C. Thrust
    D. Withdrawn
    E. Directed

As increasing dependence on information systems develops, the need for such systems to be reliable and secure also becomes more essential. As growing numbers of ordinary citizens use computer networks for banking, shopping, etc., network security is potentially a massive problem. Over the last few years the need for computer and information system security has become increasingly evident, as web sites are being defaced with greater frequency, more and more denial-of-service attacks are being reported, credit card information is being stolen, there is increased sophistication of hacking tools that are openly available to the public on the Internet, and there is increasing damage being caused by viruses and worms to critical information system resources.

At the organizational level, institutional mechanisms have to be designed in order to review policies, practices, measures, and procedures to review e-security regularly and assess whether these are appropriate to their environment. It would be helpful if organizations share information about threats and vulnerabilities, and implement procedures for rapid and effective cooperation to prevent, detect and respond to security incidents. As new threat and vulnerabilities are continuously discovered there is a strong need for cooperation among organizations and, if necessary, we could also consider cross-border information sharing. We need to understand threats and dangers that could be vulnerable to and the steps that need to be taken to mitigate these vulnerabilities. We need to understand success control systems and methodology, telecommunication and network security, and security management practice. We should be well versed in the area of application and systems development security, cryptography, operations security and physical security.

The banking sector is poised for more challenges in the near future. Customers of banks can now look forward to a large array of new offerings by banks. From an era of mere competition, banks are now cooperation among themselves so that the synergistic benefits are share among all the players. This would result in the formation of shared payment networks (a few shared ATM networks have already been commissioned by banks), offering payment services beyond the existing time zones. The Reserve Bank is also facilitating new projects such as the Multi Application Smart Card project which, when implemented, would facilitate transfer of funds using electronic means and in a safe and secure manner across the length and breadth of the country, with reduced dependence on paper currency. The opportunities of e-banking or e-power in general need to be harnessed so that banking is available to all customers in such a manner that they would feel most convenient, and if required, without having to visit a branch of a bank. All these will have to be accompanied with a high level of comfort, which again boils down to the issue of e-security.

One of the biggest advantages accruing to banks in the future would be the benefits that arise from the introduction of Real Time Gross Settlement (RTGS). Funds management by treasuries of banks would be helped greatly by RTGS. With almost 70 banks having joined the RTGS system, more large value funds transfers are taking place through this system. The implementation of Core Banking solutions by banks is closely related to RTGS too. Core Banking will make anywhere banking a reality for customers of each bank, while RTGS bridges the need for inter-bank funds movement. Thus, the days of depositing a cheque for collection and a long wait for its realization would soon be a thing of the past for those customers who would opt for electronic movement of funds, using the RTGS system, where the settlement would be on an almost instantaneous basis. Core Banking is already in vogue in many private sector and foreign banks; while its implementation is at different stages amongst the public sector banks.

IT would also facilitate better and more scientific decision making within banks. Information systems now provide decision makers in banks with a great deal of information which, along with historical data and trend analysis, help in the building up of efficient Management Information Systems, This, in turn, would help in better Asset Liability Management (ALM) which, in today's would of hairline margins, is a key requirement for the success of banks in their operational activities. Another benefit which e-banking could provide for, relates to Customer Relationship Management (CRM). CRM helps in stratification of customers and evaluating customer need on a holistic basis

A. When two rays of light meet each other after being reflected from these different surfaces, they have each travelled slightly different distances.
B. The key is that the light is bouncing off two surfaces, that of the oil and that of the water layer below it.
C. The distance the two rays travel determines which wavelengths, and hence colours, interfere constructively and look bright.
D. Because light is an electromagnetic wave, the peaks and troughs of each ray then interfere either constructively, to appear bright, or destructively, to appear dim.
6. Since the distance the rays travel changes with the angle as you look at the surface, different colours look bright from different viewing angles.

A. ABCD B. BADC C. BDAC
D. DCAB E. None of these

64. 1. Horses and communism were, on the whole, a poor match.
A. Fine horses bespoke the nobility the party was supposed to despise.
B. Communist leaders, when they visited villages, preferred to see cows and pigs.
C. Although a working horse was just about tolerable, the communists were right to be wary.
D. Peasants from Poland to the Hungarian Pustza preferred their horses to party dogma.
6. "A farmer's pride is his horse; his cow may be thin but his horse must be fat," went a Slovak saying.

A. ACDB B. DBCA C. ABCD
D. DCBA E. None of these

65. 1. Making people laugh is tricky.
A. At times, the intended humour may simply not come off.
B. Making people laugh while trying to sell them something is a tougher challenge, since the commercial can fall flat on two grounds.
C. There are many advertisements which do amuse but do not even begin to set the cash tills ringing.
D. Again, it is rarely sufficient for an advertiser simply to amuse the target audience in order to reap the sales benefit.
6. There are indications that in substituting the hard sell for a more entertaining approach, some agencies have rather thrown out the baby with the bath water.

A. CDBA B. ABCD C. BADC
D. DCBA E. None of these

Fill in the blanks with the most suitable option. (Q. 66-90)

66. The secretary _____ the V.C. informed us that the V.C. was out of station.
A. from B. to C. with
D. for E. on

67. He saw me _____ the glasses.
A. From B. across C. in
D. Through E. into

68. My brother wishes to get a degree _____ Management.
A. in B. at C. from
D. on E. within

69. Why don't you come and meet me _____ the canteen.
A. On B. in C. at
D. From E. upon

70. The boy jumped into the river and swam it _____.
A. On B. in C. at
D. From E. across

71. I can't break. It's _____ iron.
A. as hard as D. so hard than
B. so hard as E. None of these
C. as hard than

72. _____ of them knew about the plan because it was secret.
A. Some B. Any C. No one
D. None E. None of these

73. This question is _____ difficult for me.
A. so much B. too much C. too
D. enough E. None of these

74. Living here at the top of the mountain with no one else near you must be very _____.
A. sole B. alone C. only
D. lonely E. None of these

75. The tailor made him a new _____.
A. clothes B. suit C. dress
D. wear E. None of these

76. The clock _____ and we realized it was two o'clock.
A. hit B. struck C. turned
D. rang E. None of these

77. My car _____ so I had to come by bus.
A. fell down B. fell over C. broke down
D. broke up E. None of these

78. Look what Father _____ me when he came home from work.
A. brought B. took C. carried
D. fetched E. None of these

79. I _____ have coffee than tea.
A. like more to B. prefer C. would rather
E. had better E. None of these

80. I'll have to buy _____ trousers.
A. a B. two C. a pair of
D. a couple of E. None of these

81. He is rich; __, his cousin is poor.
A. Therefore B. However C. Otherwise
D. Unless E. Because

82. You'd better take a taxi. __, you'll arrive late.
A. Consequently B. Furthermore C. Otherwise
D. But E. And

83. Phil was not thirsty; __, he drank five glasses of water.
A. However B. Moreover C. Furthermore
D. But E. Since

84. The kids didn't study. _____, they failed the course.
A. Therefore B. Nonetheless C. Otherwise
D. But E. Because

85. The weather was terrible. __, we decided to delay our trip.
A. Furthermore B. Besides C. Therefore
D. Unless E. Since

86. You must buy the tickets; _____, we won't be able to see that play.
A. Otherwise B. Though C. Besides
D. Although E. Therefore

87. We live in the same building; _____, we hardly see each other.
A. However B. Therefore C. Furthermore
D. Besides E. Otherwise

88. He didn't earn enough money. __, his wife decided to get a job.
A. Moreover B. Therefore C. Although
D. But E. Because

89. That house isn't big enough for us, and __, it's too expensive.
A. Furthermore B. Hence C. Although
D. Therefore E. But

90. We have plenty of money and workers; __, we hope to finish the house remodeling soon.
A. Nevertheless B. Unless C. Thus
D. Since E. Because

SET- B University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), End Term Examination, 2016-17

Subject: English for Professional Purposes–II Subject Code: AHE 4083
Time: 90 Minutes Max. Marks: 90

VERBAL APTITUDE

In Q 1 to 45 identify the correct option

1. The inevitable consequences were difficult for him to bear.
   A. Adjective B. Noun C. Pronoun
   D. Adverb E. None of these
2. As I was very sincere and committed to my work, I was offered a week off.
   A. Conjunction B. Preposition C. Pronoun
   D. Noun E. Adverb
3. Why did Kattapa kill Bahubali?
   A. Noun B. Preposition C. Conjunction
   D. Adjective E. Adverb
4. The utterly delicious dinner didn't cost me a fortune.
   A. Adjective B. Verb C. Noun
   D. Adverb E. None of these
5. The unusually high pricing of the things couldn't deter him from making that purchase.
   A. Adjective B. conjunction C. preposition
   D. Adverb E. Noun
6. Rounding off is a common phenomenon in online shopping websites.
   A. Present participle B. Past Participle C. Gerund
   D. Infinitive E. None of these
7. In this semester, ME is offering a course on Refrigeration and refrigerating tools.
   A. Present participle B. Past Participle C. Gerund
   D. Infinitive E. None of these
8. The _____books because of no use had attracted lots of dust.
   A. Stacked B. stacking C. to stack
   D. stacken E. Both B & C
9. _____doesn't affect your eyes that much as the laptop does.
   A. read B. to read C. reading
   D. the read E. Both A & D
10. _____ or not to die is the question.
    A. dying B. to die C. dead
    D. death E. Both A & B
11. If this guy rejects our offer, we'll have to_____and start the whole recruitment process again
    A. Go back to square one D. Be starry eyed
    B. Be squeaky clean E. None of the above
    C. Be on the square
12. Some students have _____that they are in the University for the Studies.
    A. Lost sight of the fact D. Become sitting ducks
    B. Set there sight on E. None of the above
    C. In their sights
13. The country will_____on 7th July 2017.
    A. Go to the polls D. Go insane
    B. Go postal E. None of the above
    C. Go dutch
14. You may not get a better offer. I'd _____, if I were you.
    A. Have an itchy feet D. Not be just a pretty face
    B. Strike while the iron is hot E. None of the above
    C. Be the jewel in the crown
15. The prime minister has _____ in the decision making and not the parliament, in such cases.
    A. A lower hand D. Hand in hand
    B. An upper hand E. None of these
    C. No hand
16. I need to find some kind of chemical that will _____the weeds in the garden.
    A. keep off B. keep out C. keep down
    D. keep on E. keep of
17. Last week I _____ that book you wanted in a small shop.
    A. came up B. came into C. came across
    D. came by E. came upon
18. You need not worry that you'll be left on your own because I'll always _____ you.
    A. stand for B. stand to C. stand by
    D. stand from E. stand out
19. Have you _____ any other interesting product features that we could emphasize in the ads?
    A. come across B. drawn out C. gotten across
    D. made out E. think of
20. My new assistant needs to be _____ before I trust her to run an ad campaign like this one.
    A. broken down B. broken in C. broken up
    D. broken into E. broken out
21. Asteroids and comets slamming into Earth _____ Marge; she tries to remain under the protective cover of her roof as much as possible.
    A. worries B. worry C. worried
    D. used to worry E. None of the above
22. Someone—perhaps Emmanuel or Paul—_____ the right wine to serve with earthworm lasagna.
    A. knew B. will know C. knows
    D. know E. None of the above
23. These scissors _____ so dull that I'm not sure you could slice butter with them!
    A. is B. are C. has been
    D. had been E. None of the above
24. Physics _____ proven to be Jerry's easiest subject this semester. He brings Carol, the lab assistant, an oatmeal-raisin cookie, and, as his reward, she finishes his report.
    A. has B. have C. had
    D. will have E. None of the above
25. The good jewelry Dad gave Sara for graduation _____ more than the cheap leather wallet that David got as his reward for finishing high school.
    A. costs B. cost C. will cost
    D. had cost E. None of the above
26. FLOW: RIVER:: STAGNANT : _____?
    A. Rain B. Stream C. Pool
    D. Canal E. Cascade
27. Each of the following questions has a group. Find out which one of the given alternatives will be another member of the group or of that class.
    Arid, Parched, Droughty
    A. Draft B. Earth C. Dry
    D. Damp E. Windy
28. ABRASIVE: SKIN:: _____?
    A. dissolve : water D. corrosive: iron
    B. resilient: shock E. responsive: stimulus
    C. float : oil
29. APOLOGIZE: PENITENT :: _____?
    A. worsen: haughty D. appreciate: impressed
    B. countersign: venerable E. cherish: denigrate
    C. extenuate: blameworthy
30. MEANDER: DASH :: _____?
    A. throw: catch D. waft: plummet

B. hit: assault
C. sail: glide
E. walk: stroll

31. STUTTER: SPEECH :: _____?
A. blare: hearing
B. aroma: smell
C. astigmatism: sight
D. novocain: touch
E. deaf: inaudible

32. RETICENT: TALK :: _____?
A. abstains: devour
B. tasteless: savor
C. likely: conjecture
D. cranky: grumble
E. reserve: stack

33. PAIN: SEDATIVE:: _____?
A. comfort: stimulant
B. grief: consolation
C. trance: narcotic
D. ache: extraction
E. angst: anxiety

34. GRAIN: SALT:: _____?
A. shard: pottery
B. shred: wood
C. blades: grass
D. chip: glass
E. eye: sight

35. BACTERIA: DECOMPOSITON:: _____?
A volcano: eruption
B antibiotic: injection
C yeast: fermentation
D oxygen: treatment
E balm: pain

36. My father said, "All that glitters is not gold."
A. My father said that all that glitters was not gold.
B. My father advised that all that glitters is not gold.
C. My father remarked that all that glitters is not gold.
D. My father said that all that glitters is not gold.
E. None of the Above

37. Rakhi says to him, "You must come back home in time."
A. Rakhi tells him that he has to come come back home in time.
B. Rakhi tells him that he had to come come back home in time.
C. Rakhi tells him that he will have to come come back home in time.
D. Rakhi tells him that he must come come back home in time.
E. None of the above

38. The Judge said to inspector, "Call the thieves."
A. The Judge urged inspector not to call the thieves.
B. The Judge ordered inspector call the thieves.
C. The Judge requested inspector that he should call the thieves.
D. The Judge ordered inspector to call the thieves.
E. None of the above

39. They said to me, "May you live long."
A. They blessed me to live long.
B. They blessed me that I might live long.
C. They blessed me that I may live long.
D. They blessed me that may I live long.
E. None of the above

40. He said to us, "Are you going to market today?"
A. He asked us if we were going to market that day.
B. He asked us if you are going to market that day.
C. He asked us if we went to market that day.
D. He asked us if we had gone to market that day.
E. None of the above

41. A. The students who are coming from Japan are intelligent.
B. The students who come from Japan are intelligent.
C. The students who have come from Japan are intelligent.
D. The students who have been coming from Japan are intelligent.
E. None of the above.

42. A. My friends warned me that I should better be careful.
B. My friends warned me that I must better be careful.
C. My friends warned me that I would better be careful.
D. My friends warned me that I had better be careful.
E. None of the above.

43. A. The criminal did not confessed his crime even till the very end of his trail and said that he was innocent.
B. The criminal did not confess his crime even till the very end of his trail and said that he had been innocent.
C. The criminal do not confess his crime even till the very end of his trail and says that is innocent.
D. The criminal did not confess his crime even till the very end of his trail and said that he was innocent.
E. None of the above.

44. A. When we meet Mr. Chakraborty had been working in that institution for nine months.
B. When we meet Mr. Chakraborty had worked in that institution for nine months.
C. When we meet Mr. Chakraborty has been working in that institution for nine months.
D. When we met Mr. Chakraborty had been working in that institution for nine months.
E. None of the above.

45. A. Economists have predicted that the country's economic growth falls low to eight percent this year.
B. Economists have predicted that the country's economic growth fall low to eight percent this year.
C. Economists have predicted that the country's economic growth will fall low to eight percent this year.
D. Economists have predicted that the country's economic growth will have fallen low to eight percent this year.
E. None of the above.

Directions: Read the following passage carefully and answer the questions given below it.

For a business, still in a burgeoning state of development the performance of the gem and jewelry sector in exports is gratifying. On top of an over 50 percent growth during 2004-05, these exports are poised to grow at a similar rate during the current year, if the trend in the first half of the year is an indication. Data released by the Gems and Jewelry Export Promotion Council show that Gems exports at Rs. 2,363 crores during April-September 2005 were 51 percent higher than during the corresponding period of 2004. Since there is a greater acceleration in the tempo of exports in the second half of the year, it appears that the target of gems and jewelry exports, pegged at Rs. 5,000 crores are well within reach. Even more heartening is the fact that the export flow has been broad based, though the pace has been set by better showing in both jewelry and gem exports. This successful incursion is the outcome of a conscious strategy to build export infrastructure which includes the setting up of Gems and Jewelry Export Zones, technology missions and a liberal policy under which virtually the entire industry is thrown open to foreign direct investment. Some incentives like access to domestic market for these export-oriented units, based on the net value addition, also have helped the export drive. Besides, the industry had made helped the export drive. Besides, the industry had made efforts to capture new markets, laid stress on quality of the product and became alive to customer needs.

However, in the case of jewelry exports, the actual performance is behind the potential. With its endowment in terms of jewelry designers, low costs and high productivity, India can emerge as a major force in global jewelry exports by emphasizing on customizing jewelry rather than relying on standard products. At the same time, gems exports should not suffer by default. The recent data given lie to the claim that we have a leeway in jewelry designing. Over a period, this success on global marketing of our gems should facilitate system integration, and to exports of value-added gems with jewellery. As of now, what is significant is that we have carved a small niche in the global market in an industry that is the cutting edge of designing.

46. Which of the following are responsible for substantial growth in exports in gems and jewellery sector?
(a) Conscious strategy to build export infrastructure
(b) Liberal policy
(c) Government subsidy
A. All (a), (b) and (c) D. Only (a) and (b)
B. Only (b) and (c) E. None of these
C. Only (a) and (c)

47. Setting up of gems and jewellery exports zones has resulted into which of the following?
A. Production of gems has increased
B. Increase in gems exports
C. Increase in jewellery exports
D. Abundance of gems and jewellery in the domestic market
E. None of these

48. Which of the following words has the SAME meaning as the word 'heartening' as used in the passage?
A. Saddening B. Encouraging C. Illuminating
D. Satisfying E. Worsening

49. Which of the following factors have not helped in the export promotion?
(a) Building an infrastructure for export
(b) Encouraging foreign direct investment
A. Only (a) B. Only (b) C. Either (a) or (b)
D. Both (A) and (B) E. None of these

50. Which of the following statements is true in the context of the passage?
A. The growth in exports in gems and jewelry sector is expected to be 100 percent during 2005-2006 to that of 2004-05.
B. The growth in exports in gems and jewelry sector in 2005-2006 is expected to be more than that in 2004-05.
C. The growth in exports in gems and jewelry sector is expected to be 50 percent in 2005-2006.
D. The growth in exports in gems and jewelry sector is expected to be 25 percent in 2005-2006.
E. None of these

Directions: Read the following passage carefully and answer the questions given below it. Certain words are given in bold in the passage to help you locate them while answering some of the questions.

Though the Cold War has ended, selective tactics are still continuing for ensuring the military and economic dominance of developed countries. Various types of technology denial regimes are still being enforced which are now being mainly targeted against developing countries like India.

Today, we in India encounter twin problems. On one side there is a large scale strengthening of our neighbours through supply of arms and clandestine support to their nuclear and missile programmes and on the other side all efforts are being made to weaken our indigenous technology growth through control regimes and dumping of low-tech systems, accompanied with high commercial pitch in critical areas. Growth of indigenous technology and self-reliance are the only answer to the problem.

Thus in the environment around India, the number of missiles and nuclear powers are continuously increasing and destructive weapons continue to pile up around us, in spite of arms reduction treaties.

To understand the implications of various types of warfare that may affect us, we need to take a quick look at the evolution of war weaponry and the types of warfare. I am highlighting this point for the reason that in less than a century we could see change in the nature of warfare and its effects on society.

In early years of human history it was mostly direct human warfare. During the twentieth century up to about 1990, the warfare was weapon driven. The weapons used were guns, tanks, aircraft, ships, submarines and the nuclear weapons deployed on land/sea/air and also reconnaissance spacecraft. Proliferation of conventional nuclear and biological weapons was at a peak owing to the competition between the superpowers.

The next phase, in a new form, has just started from 1990 onwards. The world has graduated into economic warfare. The means used is control of market forces through high technology. The participating nations, apart from the USA, are Japan, the UK, France, Germany, certain South-East Asia countries and a few others. The driving force is the generation of wealth with certain types of economic doctrine.

The urgent issue we need to address collectively as a nation is, how do we handle the tactics of economic and military dominance in this new form coming from the backdoor? Today technology is the main driver of economic development at the national level. Therefore, we have to develop indigenous technologies to enhance our competitive edge and to generate national wealth in all segments of economy. Therefore, the need of the hour is *arm India with technology.*

51. Why do certain countries use selective tactics against developing countries?
A. To help developing countries gain military and economic independence
B. To help developing countries govern themselves and be economically independent
C. To ally with developing countries to dominate over other developed countries
D. To curtail their domination over developing countries
E. None of these

52. Choose the word which is closest in meaning as the word **Proliferation**
A. Explosion D. Abundance
B. Devastation E. Extraction
C. Discomfiture

53. Enforcement of technology denial regimes by developed countries implies which of the following?
A. Dominance of developing countries over developed ones
B. Exploitation of developing nations by the mightier ones
C. Targeting of developed countries by developing countries
D. Sympathizing with underprivileged countries
E. None of these

54. The striking difference in warfare before and after 1990 was the shift from
A. guns, tanks, etc. to nuclear weapons
B. ships and submarines to spacecrafts
C. weaponry to economic warfare
D. economic forces to high technology driven warfare
E. None of these

55. What is the general outcome of arms reduction treaties as a whole according to the author of the passage?
A. They seem to have become totally defunct
B. They have achieved the desired outcome in most cases
C. They have resulted in curbing the trade of destructive weapons
D. Piling up of weapons has significantly reduced due to such treaties
E. None of these

Identify correct option in Q 56 to 65

56. Don't make him _____ it if he doesn't want to.
A. do B. to do C. doing
D. that he do E. None of these

57. He's _____ to know the answer.
A. likely B. probable C. maybe
D. probably E. None of these

58. She came _____ because her car has broken down.
A. walking B. by foot C. with foot
D. on foot E. None of these

59. _____ him go out if he wants to.
A. Allow B. Leave C. Let
D. Permit E. None of these

60. I didn't hear what he was _____
A. telling B. saying C. talking
D. speaking E. None of these

61. I _____ watching this program because it is very interesting.
A. amuse B. please C. delight
D. enjoy E. None of these

62. That student _____ his hand every time I ask a question.
A. gets out B. gets up C. rises
D. puts up E. None of these

63. I _____ hands with him when he came in.
A. gave B. greeted C. shook
D. offered E. None of these

64. He wants to get a better ____ and earn more money.
A. job B. work C. employ
D. employment E. None of these

65. _____ the children for me while I'm out.
A. Look after B. Look to C. Take care
D. Care E. None of these

Fill in the blanks with appropriate verbal phrases:

66. He asked me to meet him _____ night.
A. from B. to C. with
D. for E. on

67. The ball stuck while passing _____ the pipe.
A. from B. across C. in
D. through E. into

68. He asked me to move _____ the window.
A. in B. at C. towards
D. on E. within

69. He asked me to meet me _____ the Canteen.
A. on B. in C. at
D. from E. upon

70. You need to go _____ the street to reach the station.
A. upon B. into C. at
D. from E. across

IN Q 71 to 76, identify the correct option which makes a coherent paragraph

71. A. Group decision making, however, does not necessarily fully guard against arbitrariness and anarchy, for individual capriciousness can get substituted by collusion of group members.
B. Nature itself is an intricate system of checks and balances, meant to preserve the delicate balance between various environmental factors that affect our ecology.
C. In institutions also, there is a need to have in place a system of checks and balances which inhibits the concentration of power in only some individuals.
D. When human interventions alter this delicate balance, the outcomes have been seen to be disastrous.
A. CBAD B. BCAD C. CABD
D. BDCA E. None of these

72. A. This very insatiability of the photographing eye changes the terms of confinement in the cave, our world.
B. Humankind lingers unregenerately in Plato's cave, still revelling, its age-old habit, in mere images of truth.
C. But being educated by photographs is not like being educated by older images drawn by hand; for one thing, there are a great many more images around, claiming our attention.
D. The inventory started in 1939 and since then just about everything has been photographed, or so it seems.
E. In teaching us a new visual code, photographs alter and enlarge our notions of what is worth looking at and what we have a right to observe.
A. EABCD B. BDEAC C. BCDAE
D. ECDAB E. None of these

73. A. With that, I swallowed the shampoo, and obtained most realistic results on the spot.
B. The man shuffled away into the back regions to make up a prescription, and after a moment I got through on the shop telephone to the Consulate, intimating my location.
C. Then, while the pharmacist was wrapping up a six-ounce bottle of the mixture, I groaned and inquired whether he could give me something for acute gastric cramp.
D. I intended to stage a sharp gastric attack, and entering an old-fashioned pharmacy, I asked for a popular shampoo mixture, consisting of olive and flaked soap.
A. DCBA B. DACB C. BDAC
D. BCDA E. None of these

74. A. But in the industrial era, if you need to destroy the enemy's productive capacity means bombing the factories which are located in the cities.
B. So in the agrarian era, if you need to destroy the enemy's productive capacity, what you want to do is burn his fields, or if you're really vicious, salt them.
C. Now in the information era, destroying the enemy's productive capacity means destroying the information infrastructure.
D. How do you battle with your enemy?
E. The idea is to destroy the enemy's productive capacity, and depending upon the economic foundation, that productive capacity is different in each case.
F. With regard to defence, the purpose of the military is to defend the nation and be prepared to do battle with its enemy.
A. FDEBAC B. FCABED C. DEBACF
D. DFEBAC E. None of these

75. A. He was so busy with them that he did not get time to eat.
B. Thousands of people came to him and asked different types of questions.
C. No one cared to see that he had his food or rest that night.
D. Swami Vivekanand once stayed in a small village.
A. BCDA B. CBAD C. DBAC
D. DBCA E. ABCD

76. A. The facts speak for themselves so, they need exposition only, not demonstration.
B. At the present moment, it is widely recognised that India holds the balance in the world-wide competition between rival ideologies.
C. It is not, of course, only in geographical sense that India is in a key position.
D. India's key position simply needs pointing out.
The proper sequence should be
A. DACB B. CDAB C. BCDA
D. BDAC E. DABC

Directions: Sentences given in each question, when properly sequenced, form a coherent paragraph. The first and last sentences are 1 and 6, and the four in between are labelled A, B, C and D. Choose the most logical order of these four sentences from among the four given choices to construct a coherent paragraph from sentences 1 to 6.

77. 1. Picture a termite colony, occupying a tall mud hump on an African plain.
A. Hungry predators often invade the colony and unsettle the balance.
B. The colony flourishes only if the proportion of soldiers to workers remains roughly the same, so that the queen and workers can be protected by the soldiers, and the queen and soldiers can be serviced by the workers.
C. But its fortunes are presently restored, because the immobile queen, walled in well below ground level, lays

eggs not only in large enough numbers, but also in the varying proportions required.

D. The hump is alive with worker termites and soldier termites going about their distinct kinds of business.

6. How can we account for her mysterious ability to respond like this to events on the distant surface?

A. BADC B. DBAC C. ADCB
D. BDCA E. None of these

78. 1. According to recent research, the critical period for developing language skills is between the ages of three and five and a half years.

A. The read-to child already has a large vocabulary and a sense of grammar and sentence structure.

B. Children who are read to in these years have a far better chance of reading well in school, indeed, of doing well in all their subjects.

C. And the reason is actually quite simple.

D. This correlation is far and away the highest yet found between home influences and school success.

6. Her comprehension of language is therefore very high.

A. DACB B. ADCB C. ABCD
D. BDCA E. None of these

79. 1. High-powered outboard motors were considered to be one of the major threats to the survival of the Beluga whales.

A. With these, hunters could approach Belugas within hunting range and profit from its inner skin and blubber.

B. To escape an approaching motor, Belugas have learned to dive to the ocean bottom and stay there for up to 20 minutes, by which time the confused predator has left.

C. Today, however, even with much more powerful engines, it is difficult to come close, because the whales seem to disappear suddenly just when you thought you had them in your sights.

D. When the first outboard engines arrived in the early 1930s, one came across 4 and 8 HP motors.

6. Belugas seem to have used their well-known sensitivity to noise to evolve an 'avoidance' strategy to outsmart hunters and their powerful technologies.

A. DACB B. CDAB C. ADBC
D. BDAC E. None of these

80. 1. The reconstruction of history by post-revolutionary science texts involves more than a multiplication of historical misconstructions.

A. Because they aim quickly to acquaint the student with what the contemporary scientific community thinks it knows, textbooks treat the various experiments, concepts, laws and theories of the current normal science as separately and as nearly seriatim as possible.

B. Those misconstructions render revolutions invisible; the arrangement of the still visible material in science texts implies a process that, if it existed, would deny revolutions a function.

C. But when combined with the generally unhistorical air of science writing and with the occasional systematic misconstruction, one impression is likely to follow.

D. As pedagogy this technique of presentation is unexceptionable.

6. Science has reached its present state by a series of individual discoveries and inventions that, when gathered together, constitute the modern body of technical knowledge.

A. BADC B. ADCB C. DACB
D. CBDA E. None of these

Fill in the blanks with right answer in Q 81 to 90

81. She's extremely rich; ___, she's not snobbish.

A. Hence B. However C. Otherwise
D. Thus E. So

82. I enjoy reading this new magazine. ___, it has good articles.

A. Moreover B. Nevertheless C. However
D. Otherwise E. Therefore

83. Jack wasn't tired. ___, he took a nap.

A. Otherwise B. Hence C. Nevertheless
D. Until E. However

84. It was a windy and rainy night. ___, I decided to go out.

A. Nevertheless B. Otherwise C. Hence
D. But E. So

85. The neighborhood isn't very interesting. I like the house, _____.

A. Moreover B. Thus C. Though
D. But E. Moreover

86. Do you think this is something ___ can be learned?

A. Who B. That C. Who
D. What E. If

87. Don't go out ___ you've finished your work.

A. As B. While C. Until
D. Because E. Since

88. Make sure you close all the windows ___ it starts to rain.

A. After B. Before C. For
D. Till E. Since

89. I think I know ___ he left the job.

A. Why B. Until C. Where
D. However E. Therefore

90. ___ I had known how much I would disappoint him, I woudn't have done it.

A. Whether B. When C. If
D. But E. Since

SET- C University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Even Semester), End Term Examination, 2016-17

Subject: English for Professional Purposes–II Subject Code: AHE 4083
Time: 90 Minutes Max. Marks: 90

VERBAL APTITUDE

Identify the correct part of speech of the underlined word from given options. (Q. 1-7)

1. The pandemonium in the hall forced my friends to leave the place.
   A. Adjective B. Noun C. Pronoun
   D. Adverb E. None of these
2. Since he was very good to me, he couldn't avoid my call.
   A. Conjunction B. Preposition C. Pronoun
   D. Noun E. Adverb
3. Where did Katappa kill Bahubali?
   A. Noun B. Preposition C. Conjunction
   D. Adjective E. Adverb
4. The utterly delicious dinner didn't cost me a fortune.
   A. Adjective B. Verb C. Noun
   D. Adverb E. None of these
5. The unusually high pricing of the things couldn't deter him from making that purchase.
   A. Adjective B. conjunction C. preposition
   D. Adverb E. Noun
6. The round off price you quoted is still higher.
   A. Present participle B. Past Participle C. Gerund
   D. Infinitive E. None of these
7. My great grandfather's reading habit is still intact.
   A. Present participle B. Past Participle C. Gerund
   D. Infinitive E. None of these

Identify the correct option to fill in the blank. (Q. 8-35)

8. _____ the higher education system a strong determination is required.
   A. To transform B. Transforming C. Transformed
   D. Transform E. Both A & B
9. An _____ room is not always better than a non-AC one.
   A. air-conditioning B. air-conditioner
   C. air-conditioned D. air-conditioneding
   E. Both B & C
10. An ___ fellow cannot make a good companion on hiking.
    A. exhausting B. exhausted C. exhauster
    D. to exhaustening E. Both B & D
11. I am in a _____situation here. Whatever I do, I incur heavy loss.
    A. No lose situation B. Win win situation
    C. No win situation D. Both A & B
    E. None of the above
12. When he started to speak in the GD, he thought he knew everything but the other members soon_____.
    A. Cut him down to size B. Cut him slack
    C. Got under his skin D. Became as silent as a grave
    E. None of the above
13. Jaipur, the pink city never fails _____on me.
    A. To be in mint condition B. To work its magic
    C. To be a mixed bag D. To be mumbo jumbo
    E. None of the above
14. It's a _____to get the building renovated before the rainy season sets in.
    A. Race against the time B. Rags to riches story
    C. Wild goose chase D. Slap in the face
    E. None of the above
15. Let me_____and see if I can come up with an answer.
    A. Become thin skinned B. Become thick skinned
    C. Put my thinking cap on D. Be on cloud nine
    E. None of the above
16. Don't worry we'll have to wait a little longer because I'm sure he'll _____soon.
    A. turn up B. turn in C. turn into
    D. turn down E. turn off
17. When spring comes, people often feel inclined to _____ their houses.
    A. do over B. do through C. do up
    D. do in E. do off
18. Has the advertising agency _____ the new promotional yet? I need it by this afternoon.
    A. dropped off B. dropped in C. dropped out
    D. dropped by E. dropped up
19. We're going to have to _____ the advertising campaign if we can't get any TV or radio time.
    A. call on B. call off C. drop off
    D. drop out E. call at
20. How do you ____ the clothes you are going to buy? How about the clothes you are going to wear for the day?
    A. pick up B. pick out C. pick on
    D. pick through E. pick upon
21. At Wendy's Washateria, all of the dryers _____ six quarters before spinning wet clothes dry.
    A. gobbles B. gobble C. gobbled
    D. will gobble E. None of the above
22. At Wendy's Washateria, each of the industrial washing machines _____ so forcefully during the spin cycle that new patrons dash outside, convinced that an earthquake is shaking the building.
    A. rocks B. rock C. rocked
    D. will rock E. None of the above
23. Everyone at Wendy's Washateria, even Myra and old Mrs. Webster, _____ that she could afford her own washer and dryer at home.
    A. wished B. have wished C. wish
    D. wishes E. None of the above
24. Every cereal bowl and casserole dish _____ slipped out of Sheila's soapy hands and shattered on the hard tile of the kitchen floor. Sheila really needs to buy a dishwasher.
    A. has B. have C. will have
    D. had E. None of the above
25. Not only fat caterpillars but also my neighbor's rabbits _____ chewed my poor tomato plants to leafless nubs.
    A. has B. have C. will have
    D. had E. None of the above
26. FLOW: RIVER:: STAGNANT : _____ ?
    A. Rain B. Stream C. Pool
    D. Canal E. Cascade
27. Each of the following questions has a group. Find out which one of the given alternatives will be another member of the group or of that class.
    Arid, Parched, Droughty, _____
    A. Draft B. Earth C. Dry
    D. Damp E. Windy
28. ABRASIVE: SKIN:: _____ ?
    A. dissolve: water B. resilient: shock
    C. float: oil D. corrosive: iron
    E. responsive: stimulus
29. APOLOGIZE: PENITENT:: _____ ?
    A. worsen: haughty B. countersign: venerable
    C. extenuate: blameworthy D. appreciate: impressed
    E. cherish: denigrate

30. MEANDER: DASH:: _____?
A. throw: catch
B. hit: assault
C. sail: glide
D. waft: plummet
E. walk: stroll

31. STUTTER: SPEECH :: _____?
A. blare: hearing
B. aroma: smell
C. astigmatism: sight
D. novocain: touch
E. deaf: inaudible

32. RETICENT: TALK:: _____?
A. abstain: devour
B. tasteless: savor
C. likely: conjecture
D. cranky: grumble
E. reserve: stack

33. PAIN: SEDATIVE:: _____?
A. comfort: stimulant
B. grief: consolation
C. trance: narcotic
D. ache: extraction
E. angst: anxiety

34. GRAIN: SALT:: _____?
A. shard: pottery
B. shred: wood
C. blades: grass
D. chip: glass
E. eye: sight

35. BACTERIA : DECOMPOSITON :: _____?
A. volcano: eruption
B. antibiotic: injection
C. yeast: fermentation
D. oxygen: treatment
E. balm: pain

Identify the most suitable form of Indirect Speech. (Q. 36-40)

36. The children said, "Let us complete homework together."
A. The children told that let them complete homework together.
B. The children told that to complete homework together.
C. The children suggested that they should complete homework together.
D. The children asked that they will complete homework together.
E. None of the above

37. Captain said, "Bravo ! We have won the match."
A. Captain exclaimed with joy that they have won the match.
B. Captain exclaimed that he had won the match.
C. Captain exclaimed with joy that they had won the match.
D. Captain exclaimed with surprise he have won the match.
E. None of the above

38. Salma said to me, "Do you have a LED TV?"
A. Salma asked me if I have a LED TV.
B. Salma asked me have you a LED TV.
C. Salma told me have you a LED TV.
D. Salma asked me if I had a LED TV.
E. None of the above

39. He remarked, "Two and two makes four."
A. He remarked that two and two would make four.
B. He remarked that two and two made four.
C. He remarked that two and two makes four.
D. He advised that two and two makes four.
E. None of the above

40. Deepak said to me, "I had finished the coffee."
A. Deepak told me that he had finished the coffee.
B. Deepak told me that he had been finished the coffee.
C. Deepak told me that he had finish the coffee.
D. Deepak told me that he finished the coffee.
E. None of the above

Pick the most appropriate option. (Q. 41-45)

41. A. You better make sure you're not late again.
B. You would better make sure you're not late again.
C. You should better make sure you're not late again.
D. You had better make sure you're not late again.
E. None of the above.

42. A. Most children liked to relax in the afternoon after they returned from school, but my daughter seems to have unlimited source of energy.
B. Most children liked to relax in the afternoon after they returned from school, but my daughter seemed to have unlimited source of energy.
C. Most children like to relax in the afternoon after they returned from school, but my daughter seems to have unlimited source of energy.
D. Most children liked to relax in the afternoon after they return from school, but my daughter seemed to have unlimited source of energy.
E. None of the above.

43. A. Leaders enable a group by imagining and create a future which will fulfil the purpose of the group.
B. Leaders enable a group by imagining and creating a future which will fulfil the purpose of the group.
C. Leaders enables a group by imagining and create a future which will fulfil the purpose of the group.
D. Leaders enable a group by imagining and will create a future which will fulfil the purpose of the group.
E. None of the above.

44. A. I remember in my youth I was used to go with my father to his shop and looked into his account books.
B. I remember in my youth I used to go with my father to his shop and look into his account books.
C. I remember in my youth I used to go with my father to his shop and looked into his account books.
D. I remember in my youth I had been used to go with my father to his shop and looked into his account books.
E. None of the above.

45. A. The principal of the school does not believe that the teachers are working hard all these years.
B. The principal of the school does not believes that the teachers are working hard all these years.
C. The principal of the school does not believe that the teachers have been working hard all these years.
D. The principal of the school does not believe that the teachers had been working hard all these years.
E. None of the above.

Read the passages and answer the questions that follow. (Q. 46-55)

Today, with a Noble prize to its credit, Grameen is one of the largest microfinance organisations in the world. It started out lending small sums to poor entrepreneurs in Bangladesh to help them grow from a subsistence living to a livelihood. The great discovery its founders made was that even with few assets, these entrepreneurs repaid on time. Grameen and microfinance have since become financial staples of the developing world. Grameen's approach, unlike other microfinanciers, uses the group-lending model. Costs are kept down by having borrowers vet one another, typing together their financial fates and eliminating expensive loan officers entirely. The ultimate promise of Grameen is to use business lending as a way for people to lift themselves out of poverty.

Recently Grameen has taken on a different challenge – by setting up operations in the US. Money may be tight in the waning recession, but it is still a nation of 1,00,000 bank branches. Globally, the working microfinance equation consists of borrowing funds cheaply and keeping loan defaults and overhead expenses sufficiently low. Microlenders, including Grameen, do this by charging colossal interest rates – as high as 60% or 70% - Which is necessary to compensate for the risk and attract bank

C. they should elect educated and experienced representatives.
D. they should look for single-party rule.
E. they should elect desirable candidates.

53. Which of the following is true in the context of the passage?
A. Good governance is related to the welfare of the people.
B. We have a mature democracy in India.
C. The leaders who change parties face hurdles in getting re-elected in our country.
D. Colonial rule was much better than the present democracy.
E. None of these

54. In a country like India, who is mainly responsible for good or bad governance?
A. The system of electing our representatives
B. The political parties
C. The voters
D. The political party in power
E. None of these

55. What is the synonym of CONCURRENCE?
A. satisfaction B. agreement C. participation
D. loyalty E. dependence

Fill in the blanks with suitable option. (Q. 56-70)

56. India's permanent envoy _____ the UN defended the country's position on NPT.
A. from B. to C. with
D. for E. on

57. When light passes _____ the prism it splits into a band of 7 colours.
A. from B. across C. in
D. through E. into

58. I do not have any interest _____ watching cricket.
A. in B. at C. from
D. on E. within

59. _____ the meeting, I fell unconscious.
A. on B. in C. at
D. from E. upon

60. Just cross_____ the bridge to get to the stadium.
A. on B. in C. at
D. from E. over

61. I gave her _____ stocking for her birthday.
A. a pair of B. a couple of C. a
D. any E. None of these

62. How _____ is your house from here?
A. long B. far C. much far
D. long way E. None of these

63. _____ he worked all day, he couldn't finish the job.
A. Although B. Even C. In case
D. In spite E. None of these

64. I'll leave him a note _____ he'll know where we are.
A. so that B. that C. in order
D. for E. None of these

65. You _____ drive carefully. The roads are wet.
A. had rather B. would rather C. had better
D. would better E. None of these

66. I wanted to go there by plane but I hadn't enough money to pay for the _____.
A. journey B. travel C. voyage
D. fly E. None of these

67. He failed the examination three times but __ he passed.
A. at the end B. at finish C. at last
D. at least E. None of these

68. Tennis is a _____ invented by an Englishman a hundred years ago.
A. game B. play C. toy
D. match E. None of these

69. He has spent a large _____ of money on his new house.
A. deal B. amount C. number
D. piece E. None of these

70. _____ at the door before you come into the room.
A. Hit B. Knock C. Touch
D. Strike E. None of these

Choose the proper sequence of sentences. (Q. 71-77)

71. A. It results from a carefully revised plan.
B. Men work together for a cause or purpose.
C. Team work does not just happen.
D. It must be clearly known to them.
A. BCAD B. CBDA C. BCDA
D. CABD E. None of these

72. A. I will give you a copy of it.
B. The book was published in New York.
C. It is a very interesting book.
D. It deals with mankind's political future.
A. DCBA B. CBDA C. BDCA
D. DBCA E. None of these

73. A. He had inherited that money from an uncle.
B. Mr. Maini, the uncle, was an innocent man.
C. It was regarding investing ten thousand rupees in my firm.
D. He agreed to my proposal.
A. DCAB B. BADC C. CDAB
D. BDCA E. None of these

74. A. His elbow was bleeding.
B. He had a few bruises on his left hand.
C. I went into the crowd and was relieved to see that he wasn't very badly injured.
D. A crowd gathered around my brother before he could stand up.
A. BCDA B. DABC C. BADC
D. DCBA E. None of these

75. A. Since then, intelligence tests have been mostly used to separate dull children in school from average or bright children, so that special education can be provided to the dull.
B. In other words, intelligence tests give us a norm for each age.
C. Intelligence is expressed as intelligence quotient, and tests are developed to indicate what an average child of a certain age can do: what a 5-year-old can answer, but a 4-year-old cannot, for instance.
D. Binet developed the first set of such tests in the early 1900s to find out which children in school needed special attention.
E. Intelligence can be measured by tests.
A. CDABE B. DECAB C. EDACB
D. CBADE E. None of these

76. A. Similarly, turning to caste, even though being lower caste is undoubtedly a separate cause of disparity, its impact is all the greater when the lower-caste families also happen to be poor.
B. Belonging to a privileged class can help a woman to overcome many barriers that obstruct women from less thriving classes.
C. It is the interactive presence of these two kinds of deprivation—being low class and being female—that massively impoverishes women from the less privileged classes.
D. A congruence of class deprivation and gender discrimination can blight the lives of poorer women very severely.
E. Gender is certainly a contributor to societal inequality, but it does not act independently of class.
A. EABDC B. EBDCA C. DAEBC
D. BECDA E. None of these

77. A. This feeling of an extensive group gives rise to a fellow feeling, a feeling of brotherhood amongst the citizens.

B. This feeling takes up beyond the bounds of family, caste, religion as well as region and helps us to develop a broad perspective that all of us together constitute an extensive group called 'the nation'.
C. National integration is the feeling among all the citizens of a country that they all are part of one nation.
D. We do not, then, limit our thinking to our own caste or religion but think about all our fellow citizens.

A. CDAB B. CABD C. CDBA
D. CBDA E. CBAD

Choose the most logical order of sentences to construct a coherent paragraph. The first and last are 1 and 6. (Q. 78-80)

78. 1. Commercially reared chicken can be unusually aggressive, and are often kept in darkened sheds to prevent them pecking at each other.
A. The birds spent far more of their time—up to a third—pecking at the inanimate objects in the pens, in contrast to birds in other pens which spent a lot of time attacking others.
B. In low light conditions, they behave less belligerently, but are more prone to ophthalmic disorders and respiratory problems.
C. In an experiment, aggressive head-pecking was all but eliminated among birds in the enriched environment.
D. Altering the birds' environment, by adding bales of wood-shavings to their pens, can work wonders.
6. Bales could diminish aggressiveness and reduce injuries; they might even improve productivity, since a happy chicken is a productive chicken.

A. DCAB B. CDBA C. DBAC
D. BDCA E. None of these

79. 1. The concept of a 'nation-state' assumes a complete correspondence between the boundaries of the nation and the boundaries of those who live in a specific state.
A. Then there are members of national collectivities who live in other countries, making a mockery of the concept.
B. There are always people living in particular states who are not considered to be (and often do not consider themselves to be) members of the hegemonic nation.
C. Even worse, there are nations which never had a state or which are divided across several states.
D. This, of course, has been subject to severe criticism and is virtually everywhere a fiction.
6. However, the fiction has been, and continues to be, at the basis of nationalist ideologies.

A. DBAC B. ABCD C. BACD
D. DACB E. None of these

80. 1. In the sciences, even questionable examples of research fraud are harshly punished.
A. But no such mechanism exists in the humanities—much of what humanities researchers call research does not lead to results that are replicable by other scholars.
B. Given the importance of interpretation in historical and literary scholarship, humanities researchers are in a position where they can explain away deliberate and even systematic distortion.
C. Mere suspicion is enough for funding to be cut off; publicity guarantees that careers can be effectively ended.
D. Forgeries which take the form of pastiches in which the forger intersperses fake and real parts can be defended as mere mistakes or aberrant misreading.
6. Scientists fudging data have no such defences.

A. BDCA B. ABDC C. CABD
D. CDBA E. None of these

Fill in the blanks with the most suitable option. (Q. 81-90)

81. He couldn't go home, _____ he had no place to go.
A. For B. But C. Therefore
D. Since E. None of the above

82. She owned a car, _____ she didn't know how to drive it.
A. Yet B. Since C. For
D. Though E. None of the above

83. He didn't want help, _____ did he ask for it.
A. Nor B. Yet C. So
D. If E. Although

84. Not only the movie __ also the play was good.
A. Since B. And C. But
D. Yet E. None of the above

85. _____ you see me, get in the car.
A. As soon as B. While C. Whether
D. Soon E. None of the above

86. My friend John, _____ lives in California, has a car with a New York license.
A. Whose B. Who C. That
D. Which E. None of the above

87. It's usually quite simple to cross the border between the USA and Canada _____ these two countries have friendly relations.
A. Yet B. And C. Because
D. Since E. None of the above

88. _____ a university professor returns from a sabbatical leave, he should be better qualified.
A. Until B. When C. Even though
D. Though E. Since

89. She's _____ a snobbish person _____ nobody likes to be with her.
A. Such...that B. Such...as C. So...that
D. Both A and C E. None of the above

90. She had to go, _____ she called a friend to drive her.
A. So B. Since C. Yet
D. But E. And

University Roll No. .....

END-TERM EXAMINATION, 2016-17

B. Tech. (Sem. IV) CSE

Discrete Mathematics (CSE – 4005)

Time: 2 Hrs. 30 Minutes **Max. Marks: 40**

Note : Attempt ALL sections. The symbols have their usual meaning.

[SECTION – A]

Note : Attempt ALL questions. **(1 x 16 = 16 Marks)**

Q.1. Find the number of elements in the power set P(s) of the set $S=\{\{\phi\},\{1\},\{2,3\}\}$

Q.2. Draw a Venn diagram of set A, B, C. Where $A \subseteq B$, sets B and C are disjoint, but A and C have elements in common.

Q.3. If $A=\{1,2,3,4\}$ &$R=\{(1,1), (2,2), (2,3), (3,2), (4,2), (4,4)\}$ is relation in A. Is R

(i) Reflexive (ii) Symmetric (iii) Transitive (iv) Antisymmetric ?

Q.4. Find the solution of the recurrence relation

$a_{n+1} - 2a_n = 5$ where $n \geq 0, a_0 = 1$

Q.5. Find the generating function for the sequence $\{1,1,1/2,1/3,1/4,....\}$.

Q.6. How many numbers greater than a million can be formed with digits 2, 3, 0, 3,4, 2,3 ?

Q.7. One of ten keys opens the door .If we try the keys one after one then, what is the probability that the door is open on

(i) The first attempt (ii) The second attempt

Q.8. Show that the proposition $(p \vee q) \wedge (\sim p) \wedge (\sim q)$ are contradiction.

Q.9. Negate the statement

"For all real numbers x, if $x > 3$ then $x^2 > 9$".

Q.10. Obtain the principal disjunctive normal form of $p \rightarrow q$.

Q.11. What is the chance of drawing 7 or 11 with two dice?

Q.12. If p and q are two statements show that the implication $p \Rightarrow q$ and its contra-positive are logically equivalent.

Q.13. Let G be a group and $a \in G$ if O (a) =17 then find $O(a^8)$.

Q.14. Index of a subgroup is 5 and it's order is 3.Then find the order of the group.

Q.15. Suppose a graph has vertices of degree 0,2,2, 3,and 9. How many edges does the graph have?

Q.16. How many vertices and how many edges do the following graph have? (i) $K_{m,n}$ (ii) C_n

Section – B

Note: Attempt Any FOUR Questions. **(3x4 = 12 Marks)**

Q.1. Show that the set of all integers $\mathbb{Z}$ forms a group with respect to the binary operation '$*$' defined by the rule

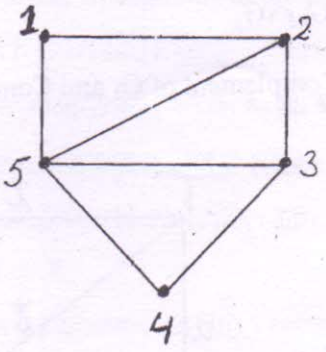
$a * b = a + b + 2 \ \forall a, b \in \mathbb{Z}$. Also find 1^3

Q.2. Prove that a necessary and sufficient condition for a non-empty subset H of a group $(G, *)$ to be a subgroup is that $a \in H, b \in H \Rightarrow a * b^{-1} \in H$, where b^{-1} is inverse of b in G.

Q.3. Prove that the order of the elements a and $x^{-1}.a.x$ are the same, where a, x are any two element of a group $(G, .)$

Q.4. Define complete graph and complete bipartite graph. Draw the complete graph k_6 and complete bipartite graph $k_{3,3}$. Find whether the k_6 and $k_{3,3}$ are isomorphic or not.

Q.5. Find the incidence matrix and adjacency matrix for the following graph:



Section – C

Note: Attempt Any THREE Questions. **(4×3 = 12 Marks)**

Q.1. Show that the set of all matrices of the form $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} : a, b \in \mathbb{Z}$ is a commutative ring with respect to matrix addition and multiplication. Does it have unity? Is it a field?

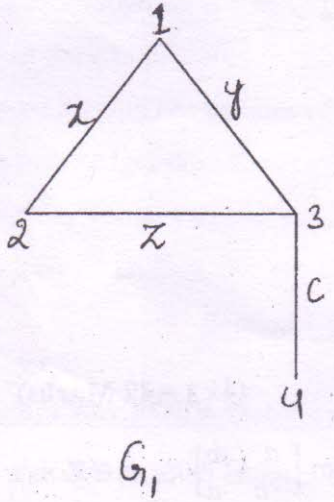
Q.2. Find all the distinct left cosets of $3\mathbb{Z}$ in$(\mathbb{Z}, +)$. Verify that these distinct left cosets partition $\mathbb{Z}$, where $(\mathbb{Z}, +)$ is the additive group of integers and $3\mathbb{Z} = \{0, \pm 5, \pm 10, \pm 15, \ldots \ldots\}$. Also, by means of an example, verify the theorem $aoH = H$ iff $a \in H$.

Q.3. Define a group homomorphism. Show that the function $f : (\mathbb{R}^+, \times) \rightarrow (\mathbb{R}, +)$ defined by $f(x) = \log x$ is a group homomorphism, where $\mathbb{R}^+$ is the set of positive real no's and $\mathbb{R}$ is the set of real no's. Also find the Kernel of this homomorphism.

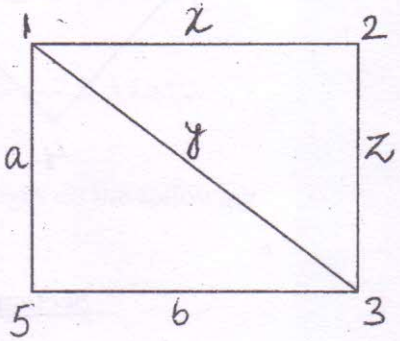
Q.4. If G_1 and G_2 are following graphs: Find graph of

(i) $G_1 \cup G_2$ (ii) $G_1 \cap G_2$

(iii) $G_1 \oplus G_2$ (iv) Complement of G_1 and Complement of G_2



G_1



G_2

END Term Examination, 2016-17

Course: B.Tech.(CSE) **Year: II** **Semester: IV**

Object Oriented Programming (CSE4004)

Uni. Roll No: .....

Time: 2½ Hrs. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. Enlist the use of JIT.

II. Enlist the need of UML.

III. Enlist the difference between link & association.

IV. Define Composition also draw the notation.

V. Why Java called as robust language?

VI. Enlist the difference between generalization & specialization.

VII. What is the difference between throw & throws?

VIII. How Applet is differing from Application?

IX. Enlist the difference between Swing components and AWT components.

X. Define Layout Manager.

XI. Draw the applet life cycle.

XII. Explain sleep() method under Thread class.

XIII. What does Class.forName() method do?

XIV. Define the directory structure for servlet programming.

XV. Enlist any one major difference between EJB and Java Bean.

XVI. Define Stub & Skeleton.

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Write the steps to create JDBC program.

II. Explain RMI using its architecture.

III. Define web service also enlist the difference between SOAP and RESTful.

IV. Explain the difference between different types of statement used in JDBC.

V. Write a program to add two integer values using RMI.

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. What is EJB? Explain different types of EJB.

II. What are servlets? Explain servlet life cycle in details with example.

III. What are database driver? Explain different types of database drivers?

IV. Write a JDBC program to read data from database using any type of database driver.

End Term Examination,
Even Semester 2016-17

Course: B. Tech. II Yr. **Univ. No.** ----------
Title of Paper: Environmental Studies
Paper Code: AHS-3001
Time: 2.5 Hours **Max. Marks**: 40

Section-A

Note: Attempt All Questions **1 X 16 = 16 Marks**

(I) Define the term biome.
(II) What is estuarine ecosystem?
(III) Why high degree of biodiversity is needed for an ecosystem?
(IV) Which layer of atmosphere shows burning of meteors?
(V) What do you mean by stratification in ecosystem?
(VI) How fluoride acts as two-edge sword?
(VII) Why afforestation is needed?
(VIII) List the electromagnetic radiations in increasing order of frequency.
(IX) What is sanitary land-filling?
(X) Define expanding age pyramid.
(XI) Which are the green house gases causing global warming?
(XII) How is the Taj Mahal affected by acid rain?
(XIII) How EIA is useful?
(XIV) What is value added environmental education?
(XV) When did air (prevention and control of pollution) act passed in India?
(XVI) What is meant by environmental awareness?

Section-B

Note: Attempt Any four Questions **3 x 4 = 12 Marks**

(I) Write roles of citizen in protection of environment.
(II) Distinguish formal and non-formal type of environmental education.
(III) Write a note on environmental economy.

(IV) What are the guidelines for excellence in environmental education?

(V) What is the scope of environmental education?

Section-C

Note: Attempt Any Three Questions **4 x 3 = 12 Marks**

(I) Explain the role of NGOs in environmental protection.

(II) Discuss the salient features of the water (prevention and control of pollution) Act 1974.

(III) What is the status of sustainable development in India? Explain.

(IV) What is LCA? Briefly describe the types, phases and uses of it.

University Roll No.....

End Term Examination(2016-17)
Subject-Introduction to Microprocessor (CSE-4002)

Course: B.Tech.(CSE) **Year: II** **Semester: IV**
Duration: $2\frac{1}{2}$ Hrs **Maximum Marks-40**

Note:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Answer all Questions. $16 \times 1 = 16$

I. What are the advantages of memory segmentation in 8086 microprocessor?

II. List the four instructions which control the interrupt structure of the 8085 microprocessor.

III. An Intel 8085 A microprocessor is operated at a frequency of 2 MHz. If the instruction STA, 2100 H is executed, then what is the instruction cycle time ?

IV. How instruction cycle, machine cycle and T-states are related?

V. How many address lines in a 4096 x 8 EPROM CHIP?

VI. If the following program starts at 0100H, the contents of accumulator when PC reaches 0109H will be
LXI SP, 00FFH
LXI H, 0107H

MVI A, 20H
SUB M

VII. Write any four differences between 8085 and 8086?

VIII. What are the various DMA data transfer schemes?

IX. What is the significance of *READ BACK COMMAND* in 8254?

X. What is the purpose of control word written to control register in 8255?

XI. What is the difference between minimum and maximum modes of 8086?

XII. An 8085 assembly language program is given below. Assume that the carry flag is initially unset. The content of the accumulator after the execution of the program is

MVI A,07H
RLC
MOV B,A
RLC
RLC
ADD B
RRC

XIII. Draw and specify the complete bit configuration of 8085 flag Register?

XIV. Draw the control word of 8255 in I/O Mode.

XV. Total number of T States required to execute an instruction LXI H,1234H are..

XVI. How vectored interrupt is different from non vectored interrupt in 8085?

Answer any four Questions. $3 \times 4 = 12$

I. Explain control word of 8255 in BSR Mode. Consider the following addresses for PA=30H,PB=31, PC=32H, CWR=33H. Write a set of instructions to set PC0 and PC2 using BSR mode.

II. List the differences between 8253 and 8254.Write the name of different operating modes of 8253/54.

III. Differentiate between Memory mapped I/O and I/O mapped I/O.Draw and explain the control word of 8255 in I/O Mode.

IV. Write a set of instructions to Interface a key to 8255 and indicate its position on LED as shown in figure 1.
If switch is closed-LED should be ON
If switch is open-LED should be OFF
Assume Address of port A, PORT B and Control word is 30H , 31H and 32H respectively.

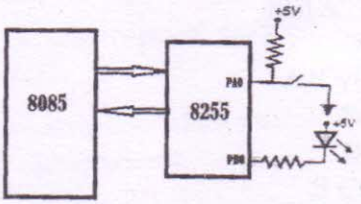


Figure 1:

V. Explain in brief different types of Data transfer scheme . Draw and explain mode set register and status register of 8237.

Answer any Three Questions. $4 \times 3 = 12$

I. Draw and explain the block diagram of Intel 8259 PIC.Also draw the flow chart for the initialization sequence of the 8259.

II. Explain the 8254 control word format and set up the 8254 as a square wave generator with 1ms period. The input frequency to the 8254 is 1 MHz. Assume counter 0 is used to generate square wave and address of counter 0=5000H and control register=50006H.

III. With the timing diagram and proper labelling explain in detail the mode 0 operation of 8253/54. Use counter 0 and count value is 05H.

IV. Draw and explain the internal block diagram of 8255 PPI. Write a set of instruction to initialize 8255 in the configuration given below

1. Port A : Output with handshake
2. Port B : Input with handshake
3. Port C_L : Output
4. Port C_U : Input
Assume address of the control word register of 8255 is 23H.

END Term Examination, 2016-17

Course: B.Tech.(CSE) **Year: II** **Semester: IV**

Computer Networks (CSE4003)

Uni. Roll No: .....

Time: 2½ Hrs. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. Explain range of TCP/IP classes?

II. What are the ranges of Pvt. IP?

III. What is Loopback address?

IV. Types of Zone in DNS?

V. Why is SMTP not used for transferring e-mail messages from the recipient's mail server to the recipient's personal computer?

VI. List the functions performed by the physical layer of 802.3 standards?

VII. Why do you require a limit on the minimum size of Ethernet frame?

VIII. In the IPv4 addressing format, what is the number of networks allowed under Class C addresses?

IX. What do understand by Manchester coding scheme?

X. We need to transmit 100 digitized voice channels using a passband channel of 20 KHz. What should be the ratio of bits/Hz if we use no guard band?

XI. What do you understate by direct and indirect delivery in routing?

XII. TCP opens a connection using the initial sequence number (ISN) 14534. The other party opens the connection with

ISN 21763. Use a message sequence diagram to show the three TCP segments during connection establishment.

XIII. Encrypt "INTERNET" using a transposition cipher with the following key:

3 5 2 1 4
1 2 3 4 5

XIV. What is difference between Leaky bucket and Token bucket algorithm?

XV. Two channels, one with a bit rate of 100 Kbps and another with a bit rate of 200 Kbps, are to be multiplexed. How this can be achieved? What is the frame rate? What is the frame duration? What is the bit rate of the link?

XVI. A voice channel occupies a bandwidth of 6 khz .We need to multiplex 12 voices channels with guard bands of 500Hz using FDM. Calculate the required bandwidth.

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. A TCP connection is using a window size of 10000 bytes and the previous acknowledgement number was 22001. It receives a segment with acknowledgement number 24001 and window size advertisement of 12000. Draw a diagram to show the situation of the window before and after.

II. Consider two TCP senders. They are at different sending hosts and go to different destinations, but pass through a common bottleneck link (that is the only bottleneck link on either of their paths). What does it mean to say that TCP provides fair sharing of bandwidth at the bottleneck link? Suppose the RTTs of the two connections are very different. Is TCP "fair" in this case? Justify your answer.

III. Is it possible for an application to enjoy reliable data transfer even when the application runs over UDP? If so, how?

IV. Suppose that the UDP receiver computes the Internet checksum for the received UDP segment and finds that it matches the value carried in the checksum field. Can the receiver be absolutely sure that no bit errors have occurred? Explain, would things be different with TCP?

V. Suppose you are sending an email from your Hotmail account to your friend, who reads his/her e-mail from his/her mail server using IMAP. Briefly describe how your email travels from your host to your friend's host. Also, what are the application layer protocols involved?

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. Explain the TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm? And Consider an instance of TCP's Additive Increase Multiplicative Decrease(AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a time out occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

II. Explain why a DNS is required with the Internet and describe its main functional parts. In relation to DNS, explain why a hierarchical naming structure is used instead of a flat structure?

III. What do you understand by symmetric and asymmetric key cryptography? If the input (32 bit) of DES round function is 10101011101010110110110110010101 then expand it to 48 bit.

IV. The following is a dump of a UDP header in hexadecimal form:06 32 00 0D 00 1C E2 17 What is the

(a)Source port number

(b)Destination port number

(c)Total length of the UDP

(d)Length of the data

(e)Considering that an IP frame can have a maximum total length of 65 535 bytes, what is the maximum length of the data in a UDP frame?

Printed Pages: 03 University Roll No.....

End Term Examination, Even Semester 2016-17
B. Tech., II-Year, IV-Semester
CSE 4001 : Operating System

Time: 2½ Hours Max. Marks: 40

Section – A

Attempt All Questions. **(1x16=16)**

I. What is dispatcher?

II. What are real time operating systems?

III. What is the main advantage of multiprogramming?

IV. Define Preemptive Kernel?

V. Mention any two properties of CPU scheduling algorithms.

VI. What is the advantage of Microkernel approach to system design?

VII. What do you understand by race-condition?

VIII. What is the use of semaphore?

IX. What are the two basic models of Inter Process Communication?

X. When will you say that the system is in safe state?

XI. What are the approaches used to handle Critical Section?

XII. What are monitors?

XIII. What is a virtual address?

XIV. Mention any two techniques used to solve the external fragmentation?

XV. Why the page sizes always powers of 2?

XVI. What is meant by thrashing?

Section – B

Attempt Any Four Questions. **(3x4=12)**

I. Consider a logical address space of eight pages of 1024 words each, mapped onto a physical memory of 32 frames.
   a. How many bits are there in the logical address?
   b. How many bits are there in the physical address?

II. Explain the various aspects of disk management briefly?

III. Given five memory partitions of 100kB, 500kB, 200kB, 300kB, and 600kB (in order), how would each of the first-fit, best-fit and worst-fit algorithms place processes of 212kB, 417kB, 112kB, and 426kB (in order)? Which algorithm makes the most efficient use of the memory?

IV. Discuss in detail about demand paging.

V. Why are segment and paging sometimes combined into one schema? Explain.

Section – C

Attempt Any Three Questions. **4 x 3 = 12 marks**

I. On a disk with 1000 cylinders, numbers 0 to 999, compute the number of tracks the disk arm must move to satisfy all the requests in the disk queue. Assume the last request serviced was at track 345 and head is moving to track 0. The queue in FIFO order contains requests for the following tracks: 123, 874,692,

475, 105, 376. Perform the computation for the following disk scheduling algorithms.

a.) FIFO b). SSTF c.) SCAN d.) LOOK

II. What is page fault? Consider the reference string 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5. There are 3 frames allotted in the memory at a time. Apply FIFO, LRU, Optimal page replacement algorithms and find the number of page faults.

III. Explain the various issues of storage allocation in a file system with that of main memory allocation.

IV. Explain the following:

a. Contiguous Allocation and Linked Allocation

b. Common schemes for defining the Logical structure of a directory.